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Question	Answer	Marks
1	**Defining the problem**	
	d is the independent variable and *V* is the dependent variable **or** vary *d* and measure *V*	**1**
	keep *A* or area (of overlap) of plates constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • circuit diagram with voltmeter connected in parallel with the capacitor • capacitor and voltmeter connected to the metal plates with no power supply in discharge part of the circuit • correct symbols for capacitor and voltmeter	**1**
	method to charge parallel plates, e.g. separate circuit diagram showing plates connected to a d.c. power supply or combined circuit with switches and d.c. power supply	**1**
	use calipers to measure *d* **or** use micrometer/calipers to measure thickness of spacers	**1**
	use rule(r) to measure lengths to determine *A* **and** *A* = length × breadth	**1**
	Method of analysis	
	plot a graph of $\frac{1}{V}$ against *d* or equivalent (e.g. *d* against $\frac{1}{V}$) (Do not accept log graphs.)	**1**
	$K = \frac{y\text{-intercept} \times C}{\text{gradient} \times A}$ or $K = \frac{C}{\text{gradient} \times AW}$ (for *d* against $\frac{1}{V}$: $K = -\frac{y\text{-intercept} \times C}{A}$)	**1**

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1	$W = \frac{1}{y\text{-intercept}}$ (for d against $\frac{1}{V}$: $W = -\frac{\text{gradient}}{y\text{-intercept}}$ or $W = \frac{\text{gradient} \times C}{AK}$)	**1**
	Additional detail including safety considerations	**6**
	D1 use gloves <u>to prevent</u> electric shock **or** do not touch metal plates to avoid shocks	
	D2 keep the <u>initial</u> p.d. across plates **or** initial charge <u>constant</u>	
	D3 method to determine the value of C, e.g. description of an experiment to measure p.d. or current against time during discharge through a resistor	
	D4 method of operation of circuit(s) using switch(es)	
	D5 description of method to fully discharge capacitor, e.g. between experiments, short-circuit the capacitor **or** use of switch in parallel with capacitor	
	D6 repeat measurements of d <u>at different points across plates</u> **and** average	
	D7 repeat measurements of V for same d **and** average V	
	D8 bottom plate resting on insulating material **or** top plate supported by strings	
	D9 use high voltage power supply to increase charge on plates **or** use a very small value of capacitance to increase voltmeter reading	
	D10 relationship valid <u>if</u> a straight line is produced (not passing through the origin)	